

ECE243 Project

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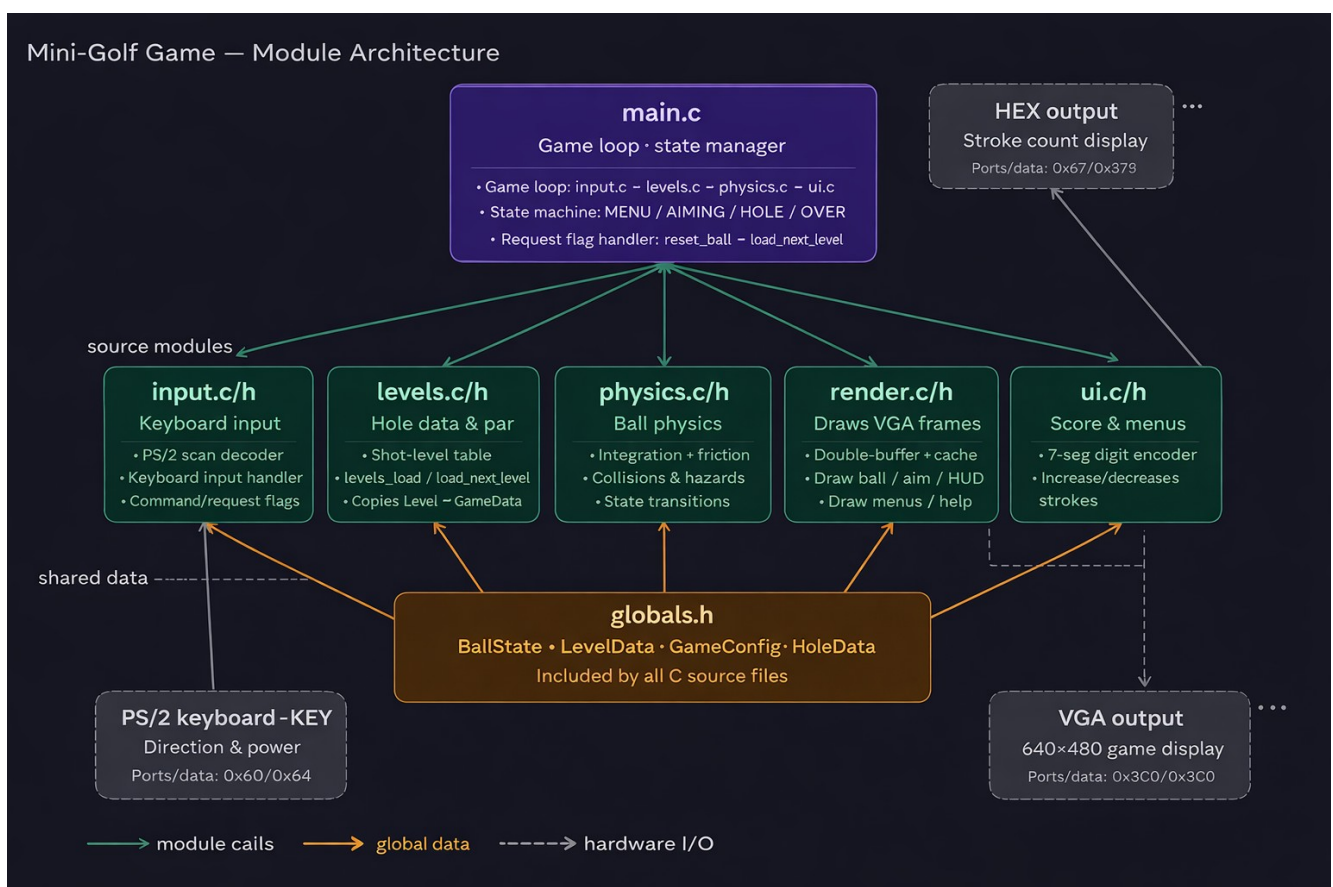
ECE243

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I. Project Description

This project implements a **Mini-Golf Game** on the DE1-SoC board. The game renders a top-down mini-golf course on a **640×480 VGA display**, where the player uses a **PS/2 keyboard** to aim and control shot power, then launches the ball toward the hole. The game features multiple levels (controlled with **KEYs**) with distinct obstacle layouts including walls and circular water hazards, realistic **2D ball physics** (velocity, friction, and collision detection), a **double-buffered VGA renderer** with a custom 5×7 bitmap font, an on-screen HUD showing stroke count, hole par, power meter, and a **HEX display** that shows the stroke count per hole, total stroke count, and level number. Game states include aiming, shooting, ball-in-motion, hole completion, and course completion screens. After completing level 3, the game transitions to a game over screen, where the total stroke count across all levels is displayed.

II. Block Diagram



III. How to Operate

Controls: (There is also a help page at main menu)

1. **← / → Arrow Keys** — Rotate aim direction left/right
2. **↑ / ↓ Arrow Keys** — Increase/decrease shot power
3. **Space / Enter** — Launch the ball
4. **R** — Reset current hole (return ball to tee)
5. **N** — Skip to next hole
6. **Enter** — Confirm / continue on result screens
7. **KEY0**—Go to main menu
8. **KEY1**—Go to level 1
9. **KEY2**—Go to level 2
10. **KEY3**—Go to level 3

Objective: Complete each hole in as few strokes as possible. The HEX display shows your current stroke count and total stroke count. Avoid water hazards; landing in one resets the ball to its previous position and adds one stroke. Sink the ball into the hole to advance to the next level. Completing level 3 ends the game and shows your final total stroke count across all three levels.

IV. Attribution Table

Xin Wang	Karan K.
Renders	Physics
UI	Input
Globals	Levels
Main	Main

The project work was completed with an approximately equal split: Xin Wang (50%) and Karan K. (50%).

Xin Wang was primarily responsible for rendering and display-related components, including the VGA rendering pipeline, UI/HUD and menu screens, HEX display logic, and shared global definitions. Within the main game loop, Xin primarily contributed to the integration of the rendering and display update flow, including the render and UI module sequencing.

Karan K. was primarily responsible for gameplay and control systems, including ball physics, collision detection, water hazard and hole logic, PS/2 keyboard and button input handling, and level structure/progression logic. Within the main game loop, Karan primarily contributed to the gameplay control flow, including input processing, reset and level-transition handling, and physics update sequencing.